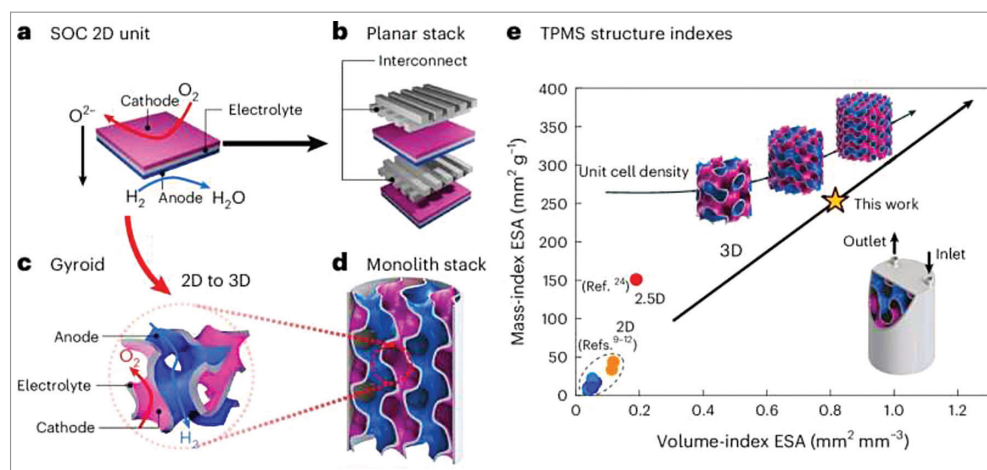


3D printed fuel cells for aerospace

Researchers at the Technical University of Denmark (DTU) have developed a 3D-printed monolithic gyroidal solid

oxide cell that delivers over one watt per gram, meeting aerospace-specific power requirements. The design is

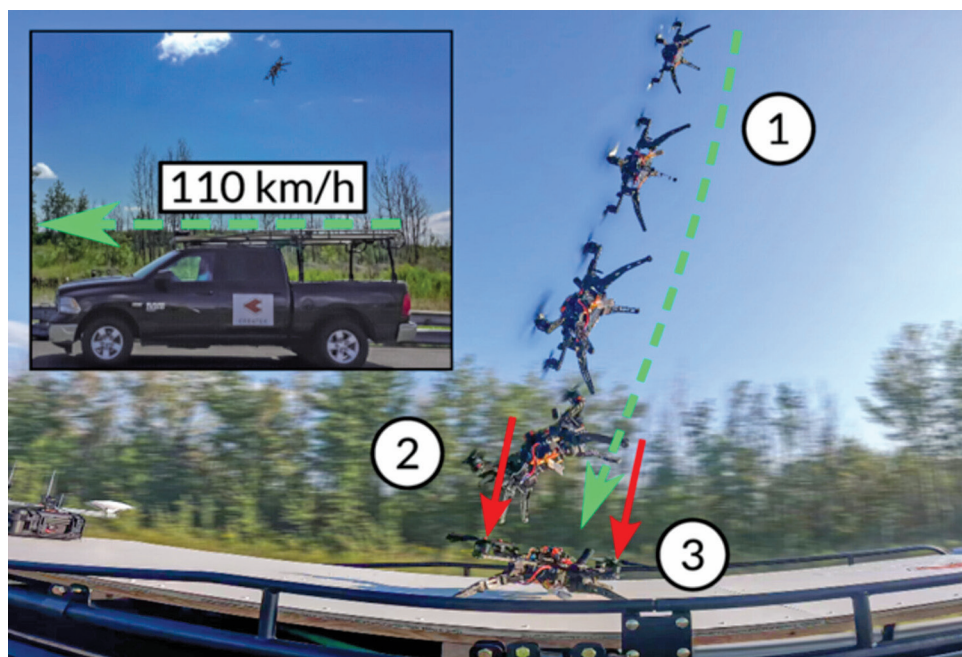


lightweight, robust, and produces hydrogen nearly ten times faster than conventional cells. Its durability under extreme conditions and simplified five-step manufacturing process drastically reduce system weight, making it a promising breakthrough for green, high-performance fuel cells in aviation and space missions.

Planar vs 3D SOC (Credit: Nature Energy, 2025)

Friction-based landing of drones on moving vehicles

Researchers at the University of Sherbrooke (Canada) have developed DART, a quadcopter using friction shock absorbers and reverse thrust to land safely on fast-moving vehicles. Drones, though common in photography, logistics, and agriculture, still face high risks during landing, which causes nearly half of the accidents. By levelling before touchdown and absorbing impact energy, DART achieves stable landings at speeds up to 110km/hr, expanding operational safety and enabling autonomous missions in dynamic, challenging environments.



Drone landing on a vehicle travelling at 110km/hr (Credit: Journal of Field Robotics, 2025)